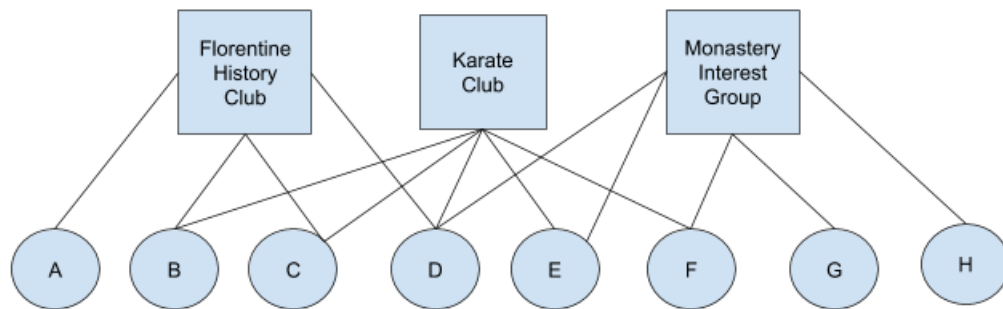


Practice Problem Set 5

1. Below is a situation where you are deciding whether to like someone, and the resulting decision that you make. Did you make this decision according to karma, status, or solidarity theories of valanced relations? (6pts)

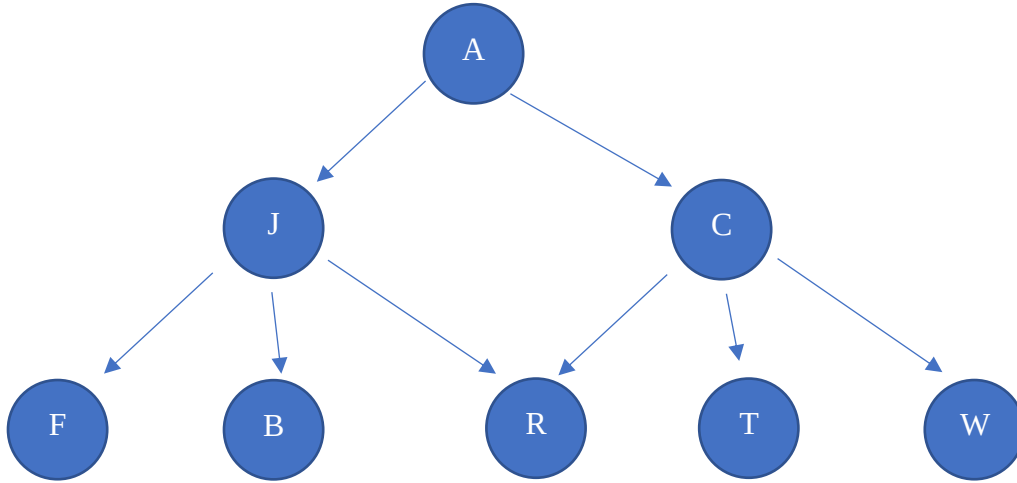
- a. A is a nice person who seems to be kind to everyone. You decide to like A.
- b. B is not liked by anyone. You decide to not like B.
- c. C is not liked by anyone. C doesn't seem so bad though. You decide to not like C.
- d. D is not liked by anyone. You are also not liked by anyone. You decide like to like D.
- e. E is a pain who doesn't like anyone. You decide to not like E.
- f. F is liked by people. You are also liked by people. You become friends. (Probably some kind of trick here that you should point out...)

2. Two-Mode Network



- Create a matrix which reflects the above graph (3pts).
- Multiply the matrix to find how many members are in each club (4pts).

3. Structural or Regular Equivalence Part 1 (6pts)



Above is a company organizational chart. Please answer whether nodes are structurally equivalent, regularly equivalent or not equivalent.

J and C:

F and B:

B and C:

A and J:

F and T:

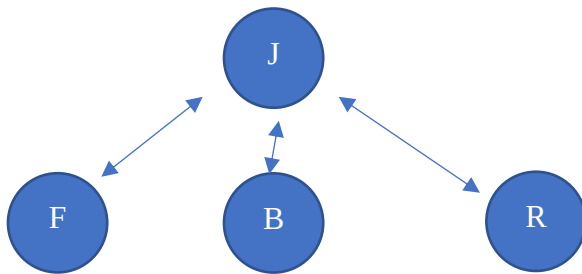
W and T:

3. Equivalence Part 2 (3pts)

Why is equivalence an important measurement concept for sociology?

What are some real-world examples of your structural equivalents? General equivalents? (2pts)

4. In the below graph, please sort the matrix into structurally equivalent blocks, including the resulting image matrix (2pts)



	B	F	J	R
B	-	0	1	0
F	0	-	1	0
J	1	1	-	1
R	0	0	1	-

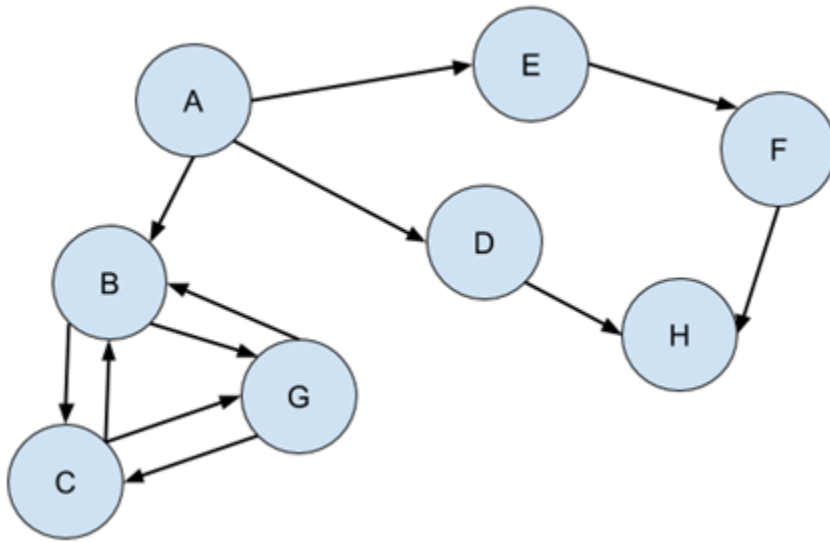
5. Assorted Questions

a) What is deference? (1pt)

b) What is the transpose of the following matrix? (1pt)

$$\begin{bmatrix} 3 & 2 \\ 2 & 5 \\ 8 & 6 \end{bmatrix}$$

6. Hierarchy and Reciprocity



a. Convert the above graph into a reachability matrix (2pts).

b. Calculate the hierarchy of the above graph (2pts)

c. Calculate the reciprocity of the above graph (2pts).

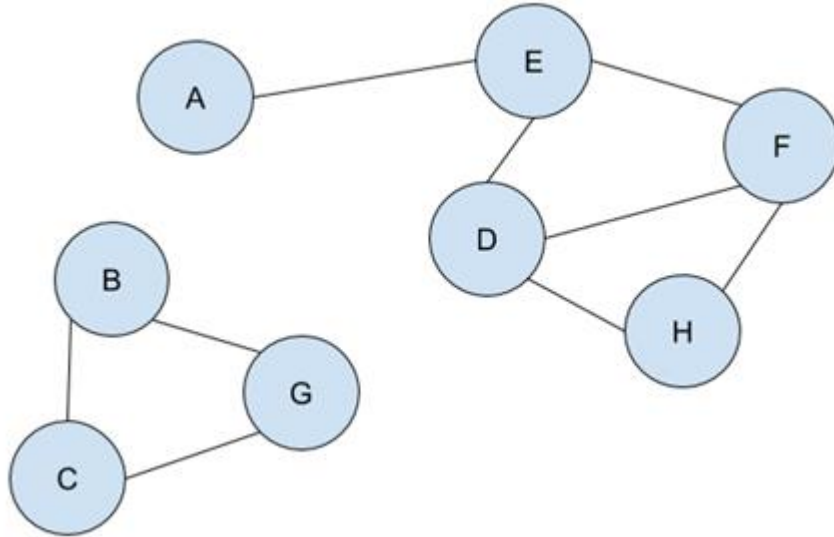
7. Matrix Multiplication (2pts each)

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \\ 3 & 2 \end{bmatrix} * \begin{bmatrix} 8 & 9 & 3 \\ 3 & 2 & 2 \end{bmatrix} =$$

$$\begin{bmatrix} 5 & 6 \\ 9 & 2 \\ 7 & 1 \end{bmatrix} * \begin{bmatrix} 3 & 5 \\ 2 & 8 \\ 3 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} 2 & 2 & 2 \end{bmatrix} =$$

8. Connectedness



	A	B	C	D	E	F	G	H
A	--	0	0	1	1	1	0	1
B	0	--	1	0	0	0	1	0
C	0	1	--	0	0	0	1	0
D	1	0	0	--	1	1	0	1
E	1	0	0	1	--	1	0	1
F	1	0	0	1	1	--	0	1
G	0	1	1	0	0	0	--	0
H	1	0	0	1	1	1	0	--

- a. Calculate the connectedness of the above graph. (2pts)

9. Multiply the adjacency matrix by itself. How many walks of length 2 start with A? (2pts)

